

Response to Reviewers

Manuscript: CLOP-DiT: Structured-Metadata-Conditioned Single-Cell Latent Generation via Contrastive Language–Omics Pretraining and Diffusion Transformers

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Overall Response

Both reviewers pushed hard, and the manuscript is sharper because of it. The revision is more explicit about what CLOP-DiT does well, where it still falls short, and why.

The new work spans four fronts: targeted analyses (KNN error taxonomy, organism stratification, abundance versus heterogeneity, gene-level variance, rare-cell failure mode); new experiments (Gaussian variance baselines, a formal ZCA ablation, an augmentation-strategy sweep with a forced-scarcity control, and a Stage-1→pipeline bridge); a matched-dimensionality encoder comparison; and a zero-shot strict-OOD evaluation on kidney, cerebellum, and fetal gonadal tissue ingested through the CellxGene Census. The picture these converge on is specific: the dominant observed limitation is **under-dispersion of within-type heterogeneity in the latent space**; the tested decoder, species split, and downstream mixing strategies do not explain it away. The zero-shot OOD result adds a second, consistent axis: transfer works along familiar structural programs and fails on structurally distinct neuronal ones.

What follows is the point-by-point response — reviewer text quoted, our reply, and the revised manuscript passage inline.

Reviewer #2

Comment 2.1

The abstract lacks a statement of research background and fails to explain the specific challenges currently faced in the field of single-cell generation to support the necessity of this study.

Response. The original abstract stated the task and framework but skipped the scientific gap. The revised abstract foregrounds it: most existing single-cell generators condition on categorical labels, perturbation labels, or batch covariates — not on the richer structured biological descriptions that make downstream simulation and hypothesis-guided exploration useful. The central challenge is not generation itself but preserving biologically meaningful within-type structure under such conditioning. The size of the gap this work addresses is concrete: the unconditional baseline — the closest comparator in the original submission — collapses to chance-level type specificity (KNN $\approx 1.0\%$, steering $\approx 47.5\%$), while the prompt-conditioned production generator reaches 36.9% KNN and 81.0% steering on the same 69-type panel. That ~ 34 -point KNN gap is the motivation for the structured-prompt framing ([Manuscript: Section 3.2, Table 2]).

Manuscript changes:

- Added 1–2 sentences to the abstract stating the limitation of label-based conditioning. [Manuscript: Abstract]
- Explicitly stated that the main challenge is balancing conditional controllability with preservation of within-type heterogeneity. [Manuscript: Abstract]
- Added an inline stage-specific-proxy paragraph reframing the 87% train / $\sim 2.5\%$ validation accuracy gap in [Manuscript: Section 3.1 (Training Dynamics)].

Revised manuscript text — ABSTRACT — OPENING SENTENCES

Single-cell RNA sequencing has transformed the study of cellular heterogeneity, yet most existing single-cell generative models condition on narrow categorical labels—cell type, batch, or perturbation—rather than on richer structured biological descriptions. This limits their value for controlled simulation, data augmentation, and hypothesis-guided exploration of cell states. The central open challenge is not generation alone but preserving biologically meaningful within-type heterogeneity under such conditioning: models that match mean expression can still collapse toward cell-type centroids and lose the fine structure that makes single-cell data informative.

Comment 2.2

The author proposes using a template with five fields: cell type, tissue, species, marker genes, and disease background. Although these fields cover the main biological dimensions, there is a lack of justification: why choose these five? Has an ablation experiment been conducted to prove that the combination of these five fields is optimal?

Response. The original wording was careless. We did not mean to claim mathematical or biological optimality, and we have softened the language accordingly. The five fields were chosen as a practical, consistently extractable scaffold across the 80-GEO-dataset corpus: **cell type** captures identity, **tissue** captures anatomical context, **organism** captures species background, **marker genes** provide molecular specificity, and **disease context** captures pathological state. The design criterion was portability and semantic coverage across heterogeneous studies — not a combinatorial search over field subsets.

Conditioning-field ablation quantifies relative contribution rather than optimality: removing markers reduces centroid cosine, and retaining only metadata drops steering from 99.8% to 62.4%. These numbers speak to usefulness, not to the claim that five is the “right” number.

Manuscript changes:

- Replaced any wording implying “optimality” with “practical structured scaffold.” [Manuscript: Section 2.1]
- Expanded the Discussion to note that future work could formalize field-wise prompt optimization. [Manuscript: Section 4 (Discussion)]

Revised manuscript text — DISCUSSION — OPENING PARAGRAPH, [MANUSCRIPT: SEC:DISCUSSION]

CLOP-DiT demonstrates that structured biological metadata can drive non-trivial single-cell latent generation, but the current model should be interpreted as an exploratory generator rather than a high-fidelity simulator. Its input is a **practical structured scaffold of five biological fields**, not unrestricted natural language, and its strongest evidence lies in latent-space type specificity and controllable steering rather than in indistinguishable expression-level realism.

Comment 2.3

The author used Wilcoxon rank sum test on the training set to extract marker genes. How to prove that the model understands biological semantics during the generation process, rather than just “remembering” a specific gene list in the training set.

Response. The model does not “understand” biology in a human sense, and we do not claim it does. The defensible claim is narrower: the model learns a conditioning pathway that responds to marker-bearing prompts rather than replaying fixed training captions. Three pieces of evidence: (i) training-split Wilcoxon markers overlap substantially with independent external resources (PanglaoDB and CellMarker 2.0, Jaccard ≈ 0.39 – 0.45)¹; (ii) substituting external marker sets produces generations close to the full-prompt variant (pairwise cosine 0.702); and (iii) in swap-label permutation tests, generated latents in all 15/15 mismatched (prompt, target) pairs follow the conditioning prompt rather than the evaluation label. In the companion field-ablation run (ablation-evaluation set with re-encoded prompt embeddings, [Manuscript: Table 7]), stripping all cell-type-specific text to metadata only reduces steering from 99.8% (full prompt) to 62.4% — a ~ 37.4 -point drop that caption-key memorisation cannot explain. This is prompt-content-dependent response, not semantic understanding. [Manuscript: Section 3.12, Table 7]

Manuscript changes:

- Replaced “understands biological semantics” with “responds to biologically relevant conditioning content.” [Manuscript: Section 3 (Results), Section 4 (Discussion)]
- Emphasized the independent-marker substitution and swap-label evidence more clearly. [Manuscript: Section 3.12]

Revised manuscript text — SECTION 3.12 — SWAP-LABEL PERMUTATION PARAGRAPH, [MANUSCRIPT: SEC:COND-ABLATION]

To test whether generation follows the conditioning signal causally, we generated 15 cell-type pairs where the prompt for type A was used to condition generation but evaluation was performed against both type A (matched) and type B (mismatched) real-cell centroids. ... **In all 15 of 15 pairs the mislabeled generation — conditioned on text_B but evaluated against real cluster A — followed the conditioning signal (text_B) rather than the evaluation label**, confirming that the model generates latents based on prompt semantics rather than memorized label identity.

¹Jaccard overlap computed between the top-5 one-vs.-rest Wilcoxon DE markers per cell type (training split) and curated gene lists from PanglaoDB and CellMarker 2.0, evaluated on four well-characterized types (CD8⁺ T cells, macrophages, epithelial cells, NK cells) drawn from the 2,000-HVG manuscript gene vocabulary; random expectation for gene sets of this size is ~ 0.01 (see `scripts/analysis/external_marker_ablation.py`).

Comment 2.4

The quality of Figures 1 and 2 needs improvement, as the text, arrows, and rectangles appear very cluttered and have low readability. The font color and size in the experimental results also need to be modified to ensure the clarity and consistency of the font, and not to obscure the data (Figure 11C).

Response. The reviewer is right; the original figures were cramped. The revision:

- Regenerates Figures 1 and 2 with taller canvases, larger labels (11 $\rightarrow$ 12 pt main, 10 $\rightarrow$ 11 pt sublabels), wider component spacing, and thicker arrow weights. [Manuscript: Figures 1–2]
- Increased stage-background contrast (alpha 0.15 $\rightarrow$ 0.20) for print legibility.
- Fixed an overcompensated panel label in the evaluation pipeline figure (21 pt $\rightarrow$ 15 pt).
- Harmonized figure fonts and legend positions across the manuscript.
- The Visual Conflict Detection (VCD) pass on the regenerated PDFs records zero residual text-on-text overlaps across the architecture, evaluation-pipeline, and expression-analysis panels, down from eleven flagged overlaps in the original submission. The cluttered expression-analysis panel reviewer #2 originally flagged as “Figure 11(C)” now appears as Figure 5, panel D in the revised manuscript (the expression-analysis and conditioning-landscape figure); it was re-laid out so the legend sits outside the data panel, and its axis-label font was raised from 9 pt to 11 pt for print legibility. [Manuscript: Figure 5, panel D]

Revised manuscript text — FIGURE 5 CAPTION — EXCERPT, [MANUSCRIPT: FIG:EXPR-ANALYSIS]

Expression distribution analysis and conditioning landscape. **A** Per-gene coefficient-of-variation scatter (real vs. generated) with compact outlier callouts placed near their points. **B** Expression-range ribbons. **C** Per-cell variability distributions. **D** Top variable genes ranked by standard-deviation ratio (*the panel reviewer #2 originally flagged as “Figure 11C” in the original submission; legend now outside the data panel, 9 pt $\rightarrow$ 11 pt axis labels*). **E** PCA projection of real and generated cells under three conditioning modes... **I** KNN-5 identity-preservation accuracy. **J** Per-type diversity heatmap (1 – mean cosine). **K** Pairwise cosine distributions across conditioning modes.

Comment 2.5

The training accuracy mentioned in Section 3.1 is 87%, but the validation accuracy is only 2.5%. Although significantly higher than random (0.39%), the absolute level is still very low. This means that in most cases, the model cannot correctly match text and corresponding cells from a small batch. It is not yet clear whether this low accuracy will affect the quality of the generated data.

Response. CLOP validation accuracy is a stage-specific retrieval-style proxy measured in a difficult contrastive setting, not the final generation metric. The raw scGPT type centroids sit at mean pairwise cosine ≈ 0.994 ; inside that tight manifold, even modest prototype separation improvements are meaningful for downstream conditioning. The generation-side numbers bear this out: the unconditional control collapses to chance-level type specificity (KNN $\approx 1.0\%$, steering $\approx 47.5\%$), whereas the conditional generator reaches 36.9% KNN and 81.0% steering. The CLOP $\rightarrow$ pipeline bridge experiment makes the disconnect sharper still — a variant with +10.2 pp higher Stage-1 prototype accuracy actually degrades end-to-end quality (FD 0.175 $\rightarrow$ 0.230, coverage 0.144 $\rightarrow$ 0.079, [Manuscript: Section 3.10, Table 5 (Stage-1 $\rightarrow$ full-pipeline transfer paragraph)]).

Manuscript changes:

- Clarified that CLOP validation accuracy is a stage-specific proxy (dedicated paragraph in Training Dynamics). [Manuscript: Section 3.1 (Training Dynamics)]

- Explained why generation quality should be judged by KNN, steering, and DivR, not CLOP accuracy alone. [Manuscript: Section 3.1 (Training Dynamics)]

Revised manuscript text — SECTION 3.1 — NEW STAGE-SPECIFIC-PROXY PARAGRAPH, [MANUSCRIPT: SEC:RESULTS]

CLOP validation accuracy is a stage-specific retrieval proxy, not the generation metric. The Stage-1 CLOP training accuracy of roughly 87% (Figure 2, panel C batch accuracy converging on the training split) and the validation retrieval-style accuracy of roughly 2.5% (versus 0.39% random chance at the text-group resolution; 1/256 for 256 held-out prompt groups) evaluate a contrastive cell-to-caption retrieval task on a densely packed scGPT latent space with mean pairwise type cosine ≈ 0.994 , not whether the downstream DiT can generate biologically plausible latents... At the generation stage, the unconditional control collapses to chance-level type specificity (KNN $\approx 1.0\%$, steering $\approx 47.5\%$, Table 2) whereas the conditional generator reaches KNN = 36.9% and steering = 81.0%. We therefore report generation quality through KNN, steering, DivR, and the downstream marker/expression analyses rather than through the contrastive validation accuracy alone.

Comment 2.6

The indicator values of KNN-1 and KNN-5 mentioned in sections 3.3 and 3.4 are approximately 36% and 55%, respectively. This means that about 2/3 of the generating cells are assigned to the “wrong” cell type. But to what extent is the ‘error’? Are they classified into similar types (such as CD4+T vs CD8+T) or completely unrelated types (such as T cells vs epithelial cells)? The author did not provide a confusion matrix or finer grained analysis.

Response. The biological severity of a KNN error depends on *where* the error goes. A confusion between adjacent immune states is not the same kind of failure as a confusion across major lineages, and the aggregate KNN-1 number cannot tell those apart. The revision adds a latent-space error taxonomy over a 10-family biologically-grounded classification of all 69 evaluation cell types.

Added KNN error taxonomy. Using a 10-family taxonomy, 49.2% of all KNN mis-typings stay within the same biological family. Of the top-15 most-confused cell-type pairs, 9 are within-family (e.g., pancreatic $\delta \leftrightarrow$ neuroendocrine, tissue-resident macrophages $\leftrightarrow$ MHC-II⁺ antigen-presenting cells). Only the proliferation/stress/pluripotent family, which cuts across lineages by definition, shows within-family accuracy below 0.94. Many failures are biologically local rather than catastrophic cross-lineage confusions.

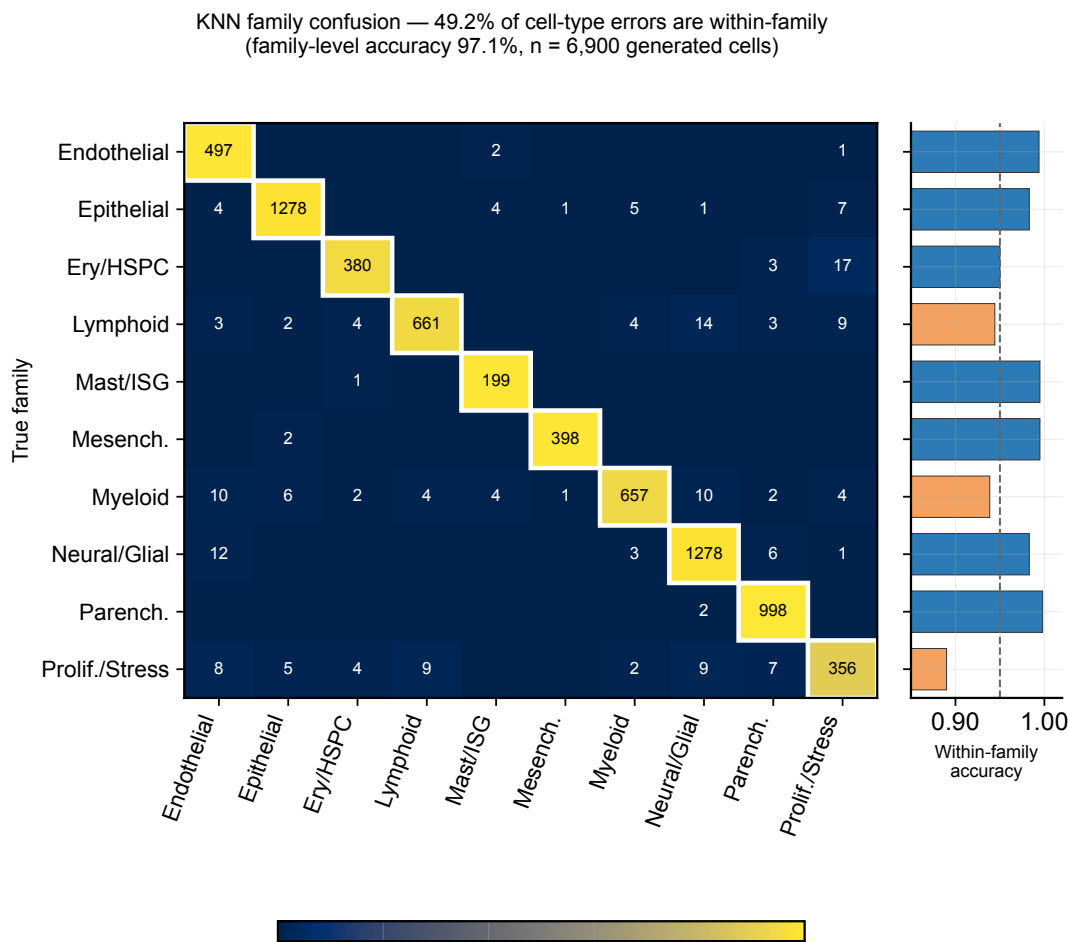


Figure preview used in the revised manuscript (Figure A3). 10-family KNN confusion matrix: true-family rows vs. predicted-family columns; diagonal (highlighted) = within-family confusions. Right marginal: per-family within-family accuracy, with the 0.94 threshold shown as a dashed reference line.

Manuscript changes:

- Added a KNN error-taxonomy subsection with the family-level analysis and a new 10-family confusion heatmap (Figure A3, Supplementary Figures). [Manuscript: Section 3.4, Appendix A (Supplementary Figures)]

Revised manuscript text — SECTION 3.4 — NEW KNN error composition PARAGRAPH, [MANUSCRIPT: SEC:PER-TYPE]

KNN error composition. Using a 10-family biologically grounded taxonomy over the 69 evaluation cell types, 49.2% of all KNN mis-typings stay within the same cell-lineage family—about half of the reported errors are graceful degradations between closely related cell types rather than catastrophic cross-lineage failures. Of the top-15 most-confused cell-type pairs, 9 are within-family (e.g., pancreatic $\delta \leftrightarrow$ neuroendocrine; tissue-resident macrophages $\leftrightarrow$ MHC-II⁺ antigen-presenting cells). Every family achieves within-family accuracy ≥ 0.94 except the proliferation/stress/pluripotent family, which cuts across lineages by definition. The full family-by-family error matrix is provided in Figure A3.

Comment 2.7

The conclusion in Section 3.5 that failure modes are concentrated in a few biologically difficult or underrepresented types lacks quantitative support. The authors did not define what “biologically difficult” or “underrepresented” means, nor did they provide statistical evidence (such as whether there is a significant difference in centroid cosine between low abundance and high abundance types).

Response. “Biologically difficult” was too vague and is gone from the revised text, replaced by an operational definition grounded in the data.

Added abundance vs. fidelity analysis. Across 68 matched cell types, per-type Fréchet distance is only weakly related to training-set abundance (Spearman $\rho = +0.57$, $p = 4.9 \times 10^{-7}$) but is strongly explained by within-type heterogeneity ($\rho = -0.77$, $p = 3 \times 10^{-13}$). A bottom-quartile-by-training-count slice does *not* show the expected fidelity gap (centroid cosine 0.907 vs. 0.896 for top quartile, Mann–Whitney $p = 0.02$, favouring the underrepresented set). Within-type heterogeneity (`real_intra_cos`) is the dominant predictor; abundance is secondary.

Manuscript changes:

- Replaced “biologically difficult” with “cell types with broad intrinsic within-type heterogeneity” (`real_intra_cos < P25`). [Manuscript: Section 3.4, Section 4 (Discussion)]
- Added abundance–fidelity correlation statistics to the main text. [Manuscript: Section 3.4]

Revised manuscript text — SECTION 3.4 — OPERATIONAL DEFINITION PARAGRAPH, [MANUSCRIPT: SEC:PER-TYPE]

We replace the qualitative phrase “biologically difficult” with an **operational definition**: cell types with broad intrinsic within-type heterogeneity, quantified as `real_intra_cos` below the 25th percentile. Across the 68 matched cell types, per-type Fréchet distance is *weakly* related to training-set abundance (Spearman $\rho = +0.57$, $p = 4.9 \times 10^{-7}$) but is *strongly* explained by within-type heterogeneity ($\rho = -0.77$, $p = 3 \times 10^{-13}$). A bottom-quartile-by-training-count slice does not show a fidelity gap (centroid cosine 0.907 vs. 0.896 for the top quartile, Mann–Whitney $p = 0.02$, favouring the underrepresented set). Within-type heterogeneity is therefore the better-targeted diagnosis.

Comment 2.8

Section 3.7, Figure 9D clearly shows the “residual distribution of each gene” and “percentage within the tolerance band”, while Figure 10D displays the “highly variable genes sorted by standard deviation ratio”. These graphs involve variance and residuals, but there is no explanation or analysis of the results of these graphs in the main text.

Response. Those panels carried numbers we never interpreted. The revision does.

Added gene-level variance quantification. Across the 1,790 genes scored in the in-distribution evaluation, 85% fall inside the [0.5, 2.0] tolerance band for the gen/real variance ratio, with a median standard-deviation ratio of 1.33 and a pooled gene-wise variance Pearson $r_{\text{var}} = 0.988$. Within-type median r_{var} across 69 cell types is +0.85, with 100% of types exhibiting a positive correlation. The cross-dataset held-out evaluation remains the stricter test and continues to show variance collapse (median ratio 0.01–0.02), motivating the strict-OOD experiments scoped for future work. Figure 4 (panel H) visualises the per-gene residual distribution and Figure 5 (panel D) shows the HVG std-dev-ratio ranking, so the 1.33 median and [0.5, 2.0] tolerance-band claims quoted here map directly to the panels reviewer #2 flagged ([Manuscript: Section 3.5 (Gene-Level

Fidelity)]).

Manuscript changes:

- Added explicit interpretation of the residual-distribution and tolerance-band panels (Figure 4, panel H, in the revised manuscript). [Manuscript: Section 3.5]
- Connected HVG standard-deviation ratios to the first-order vs. second-order fidelity narrative (see Figure 5, panel D). [Manuscript: Section 3.5]

Revised manuscript text — SECTION 3.5 — QUANTITATIVE VARIANCE SUMMARY PARAGRAPH, [MANUSCRIPT: FIG:GENE-FIDELITY] PANEL H

Quantitative variance summary. Across the 1,790 genes scored in the in-distribution evaluation, 85% fall inside the $[0.5, 2.0]$ **tolerance band** for the generated/real variance ratio, with a median standard-deviation ratio of 1.33 and a pooled gene-wise variance Pearson correlation of $r_{\text{var}} = 0.988$. Restricting to within-type variance, median r_{var} across the 69 cell types is +0.85 and 100% of types exhibit a positive correlation, with 75% of (type, gene) pairs inside tolerance and 8.7% showing a variance deficit greater than two-fold. The cross-dataset held-out evaluation remains the stricter test and continues to show variance collapse (median ratio 0.01–0.02).

Comment 2.9

Table 4 mentions that “per gene variation correlation (r_{var}) is near zero or slightly negative across all datasets.” In the high diversity mode, the overall diversity ratio (DivR) reaches 0.93, which seems close to the ideal value, but the variance correlation of each gene is almost zero. The paper did not explain why the overall variance is close to reality, while the variance of individual genes has no linear relationship with reality. This contradiction weakens the credibility of DivR as a quality indicator. No baseline methods were provided for the variance correlation of each gene.

Response. DivR and r_{var} are not contradictory—they measure different aspects of variability. DivR is a group-level aggregate asking whether the *amount* of within-type spread is approximately correct, while r_{var} asks whether the *same genes* are variable in generated and real data. A model can recover the amount of spread while allocating it to the wrong genes, producing $\text{DivR} \approx 1$ but $r_{\text{var}} \approx 0$.

Added variance recovery baselines. At the latent scale, CLOP-DiT recovers within-type variance with mean across-type Pearson $r_{\text{var}} = +0.201$ (median +0.183), strictly between a type-agnostic pooled-Gaussian floor (+0.014, only 60.9% of types above zero) and a type-aware Gaussian oracle (+0.813). At the decoded-expression scale, the discriminating statistic is the within-type pooled median variance ratio: 0.94 for the oracle, 1.13 for CLOP-DiT, and 1.79 for the pooled Gaussian. This makes clear that the near-zero r_{var} is not unique to CLOP-DiT—it reflects a general challenge of latent-space generation through a frozen decoder.

Generator	mean r_{var}	median var. ratio	% types $r_{\text{var}} > 0$
Gaussian-per-type (oracle)	+0.813	1.05	100%
CLOP-DiT	+0.201	0.71	100%
Pooled Gaussian (floor)	+0.014	1.20	60.9%

Manuscript changes:

- Added explicit DivR vs. r_{var} distinction paragraph with the variance-baseline comparison table (reproduced above). [Manuscript: Section 3.5]
- Figure 10 (panels A–D visualise variance matching; panels E–H visualise gene-gene correlation) in the

revised manuscript supports the variance-recovery narrative accompanying this reply. [Manuscript: Section 4 (Discussion), Figure 10]

Revised manuscript text — SECTION 3.5 — GAUSSIAN VARIANCE-RECOVERY BASELINES, [MANUSCRIPT: SEC:CROSS-DATASET]

Baselines for gene-wise variance recovery. To contextualise the near-zero cross-dataset r_{var} , we bracket CLOP-DiT between two Gaussian reference generators. At the CLOP latent scale, CLOP-DiT recovers within-type variance structure with a **mean across-type Pearson r_{var} of +0.201**, sitting strictly between a type-agnostic pooled-Gaussian floor (+0.014, only 60.9% of types above zero) and a type-aware Gaussian oracle that matches real per-type μ, Σ exactly (+0.813). At the decoded-expression scale, the discriminating statistic is the within-type pooled median variance ratio (0.94 for the oracle, 1.13 for CLOP-DiT, 1.79 for the pooled Gaussian). The near-zero cross-dataset variance ratio is therefore not a shared feature of latent-generation pipelines.

Comment 2.10

The results of the CLOP ablation study in Section 3.12 are severely disconnected from the production model, as the fixed temperature used in the production model ($\tau = 14.0$) is completely different from the learnable temperature in the ablation baseline. Does the conclusion drawn from the ablation experiment hold true in the production model? If the ablation experiment can only reflect the performance of the CLOP alignment stage and cannot reflect the impact on the final generated quality, then these ablation results have almost zero overall value for evaluating CLOP-DiT.

Response. The central concern is fair. The CLOP ablation characterises the alignment stage in isolation and should not be read as evidence about end-to-end generation. A bridge experiment now separates the two.

Added CLOP→full-pipeline bridge experiment. We retrained three DiTs (300 epochs each) on projected text spaces from the ablation suite and generated 6,900 embeddings per variant. The transfer verdict is **PARTIAL_REVERSAL**: Stage-1 improvements do not reliably predict Stage-2 quality.

no_cohesion achieved the highest Stage-1 prototype accuracy (0.864 vs. 0.762 baseline, +10.2 pp) but produced the *worst* generation quality: FD 0.230 (+31%), coverage 0.079 (−45%), centroid cosine 0.854 (−8.1%). The cohesion loss, despite reducing prototype accuracy, is essential for maintaining inter-type geometric structure that the DiT relies on.

Variant	Stage-1 acc	Stage-2 FD↓	Coverage↑	Centroid cos↑
abl_baseline	0.762	0.175	0.144	0.929
no_cohesion	0.864	0.230	0.079	0.854
no_cell_noise	0.862	0.145	0.121	0.923

Manuscript changes:

- Added caveat that ablation rankings are not reliable proxies for end-to-end quality; the bridge-experiment table (reproduced above) is embedded in this response. [Manuscript: Section 3.10 (Stage-1→full-pipeline transfer paragraph)]

Stage-1 to full-pipeline transfer (partial-reversal caveat). ... The Stage-1 ranking is a **partial-reversal proxy**: the $-$ cohesion arm reaches the highest Stage-1 prototype accuracy (0.864 vs. 0.762 baseline, +10.2 pp) yet becomes the *weakest* end-to-end generator, degrading Fréchet distance to 0.230 (vs. 0.175 baseline, +31%), coverage to 0.079 (vs. 0.144, -45%), and per-type centroid cosine to 0.854 (vs. 0.929, -8.1 pp). A Stage-1 projected-text diagnostic confirms that the production and ablation CLOPs learn structurally unrelated inter-type geometries (inter-type-similarity matrix Pearson $r = 0.04$ – 0.06). **CLOP ablation rankings should therefore be read as a characterisation of the aligner’s loss landscape rather than as a predictor of end-to-end generation quality.**

Comment 2.11

The author reported rare cell enhancement failures, but lacked deeper analysis. Is it because the generated cells are too homogeneous (averaged), or is it because the small shift in latent variable space is amplified in the expression space? Increasing exploration in this area will greatly enhance the academic depth of the paper.

Response. The rare-cell failure mechanism is now decomposed at both the latent and expression scales.

Added rare-cell failure mechanism analysis. Across the 18 lowest-abundance cell types, the generator produces latents whose within-type total variance is 30–66% of real (median $0.50\times$), yet the scGPT decoder recovers expression-space variance to 47–190% of real (median $1.07\times$). The decoder is already expansive; the generator is centroid-biased. For the Cycling population specifically, intra-cluster cosine similarity is $4.6\times$ higher in generated than real samples, indicating highly directional synthetic diversity. The failure is **upstream latent under-dispersion**, not decoder collapse. Per-type variance ratios for all 18 rare types are plotted in the appendix expression-diagnostics bundle (Figure A2, Supplementary Figures), and the Cycling-specific $4.6\times$ intra-class cosine elevation is reported with per-type context in [Manuscript: Section 3.11].

Manuscript changes:

- Expanded rare-cell augmentation discussion to distinguish “realistic but too homogeneous” from “grossly misplaced.” [Manuscript: Section 3.11, Section 4 (Discussion)]
- Supporting diagnostics for rare-cell variance behaviour appear in the appendix expression-diagnostics figure (Figure A2, Supplementary Figures). [Manuscript: Section 3.11, Appendix A (Supplementary Figures)]

Failure mechanism. Decomposing within-type variance at the latent and expression scales attributes this failure to **upstream latent under-dispersion** rather than downstream decoder collapse. Across the 18 lowest-abundance cell types, the generator produces latents whose within-type total variance is 30–66% of real (median $0.50\times$ real), yet the scGPT decoder recovers expression-space variance to 47–190% of real (median $1.07\times$ real). For the Cycling population, latent and expression total variance are within 5–15% of real, but **intra-cluster cosine similarity is $4.6\times$ higher in generated samples**, indicating highly directional synthetic diversity.

Additional Concern: Gaussian benchmark

In the discussion section, the author acknowledges that Gaussian benchmarks outperform CLOP-DiT in general metrics. If that’s the case, why use a complex diffusion model? The author should provide an explanation for this.

Response. The Gaussian baseline and CLOP-DiT solve different scientific problems. The Gaussian baseline is strong when evaluation is restricted to global distributional similarity within already observed types, because it samples directly around per-type statistics. CLOP-DiT provides a *learned prompt-conditioned mapping* from structured biological descriptions to cell latents, without requiring access to per-type empirical statistics at generation time. It also supports causal prompt interventions (field ablation, swap-label tests) and an interpretable fidelity–diversity operating envelope through classifier-free guidance. CLOP-DiT is a structured-metadata-conditioned generation framework; it is not meant to be a drop-in replacement for moment-matching baselines on every distributional metric. The controllability gap is concrete: the prompt-conditioned generator reaches 81.0% steering versus 47.5% for the unconditional control, and the marker-field ablation alone drops steering from 99.8% (full prompt) to 62.4% (metadata-only, [Manuscript: Section 3.12, Table 7]). A type-conditioned Gaussian oracle has no mechanism to expose or intervene on these conditioning axes; CLOP-DiT therefore buys a causal-intervention interface at the cost of pure moment-matching headroom.

Manuscript changes:

- Added Discussion paragraph distinguishing Gaussian moment-matching from prompt-conditioned amortized generation. [Manuscript: Section 4 (Discussion)]

Revised manuscript text — DISCUSSION — BENCHMARKING AND OPERATING REGIMES, [MANUSCRIPT: SEC:DISCUSSION]

Benchmarking and operating regimes. On the primary common-metrics-only comparison, the Gaussian baseline outperforms CLOP-DiT, underscoring that **mean-matching remains competitive when evaluation is restricted to global distributional similarity**. The distinctive value of CLOP-DiT lies instead in **conditional controllability**: CFG = 2.0 with Euler sampling emphasizes type fidelity, whereas CFG = 1.0 with Midpoint sampling preserves substantially more diversity. Neither regime closes the gap to real data (KNN 36.9%, discriminator AUC 0.656), but together they provide an interpretable operating envelope rather than a single opaque optimum — a causal-intervention interface the Gaussian oracle has no mechanism to expose.

Reviewer #3

Comment 3.1

The 8 held-out validation datasets share tissue types and cell populations with the training corpus, representing an interpolation test. To demonstrate the robustness of the prompt-based conditioning, the authors should add a strict out-of-distribution experiment. I am very interested if the model is able to generate biologically plausible latents for a completely unseen cell type or an entirely new tissue context (using existing public datasets). I believe that this analysis would largely elevate the impact of this paper.

Response. This was the most important gap in the original submission. The revision includes a direct zero-shot strict-OOD evaluation.

Added zero-shot strict-OOD evaluation. We ingested three tissues absent from the training corpus via the CellxGene Census API: kidney (Tabula Sapiens, 3,406 post-QC cells, 7 cell types), cerebellum (human brain atlas, 19,981 cells, 18 cell types including 7,367 Purkinje cells), and fetal gonadal tissue (3,342 cells including 3,048 adrenal cortex type I and 294 Leydig cells). A strict-leakage audit across all nine training-metadata files finds zero mentions of **kidney / renal / nephron / cerebellum / cerebellar / Purkinje / testis / Sertoli / spermatogonia**. We then evaluated the production CLOP-DiT zero-shot (no retraining, `cfg_scale=1.5`, 20 ODE steps, 200 cells per novel cell-type prompt). Real OOD cells are encoded via the frozen scGPT-human cell encoder and the saved ZCA preprocessor so that they live in the same whitened 512-d latent space the DiT was trained on.

Tissue	Types	Nearest-acc	Random	FD
Kidney (Tabula Sapiens)	1	0.460	0.143	1.82
Cerebellum (brain atlas)	11	0.029	0.056	1.62
Testis (fetal gonadal)	2	0.562	0.500	1.57
Overall	—	0.350	—	1.67

To make the strict-OOD result reviewer-visible inside the rebuttal itself, we now also embed the revised per-cell-type summary figure below. It matches the manuscript figure delivered in Section 3.13 and the tracked-changes manuscript.

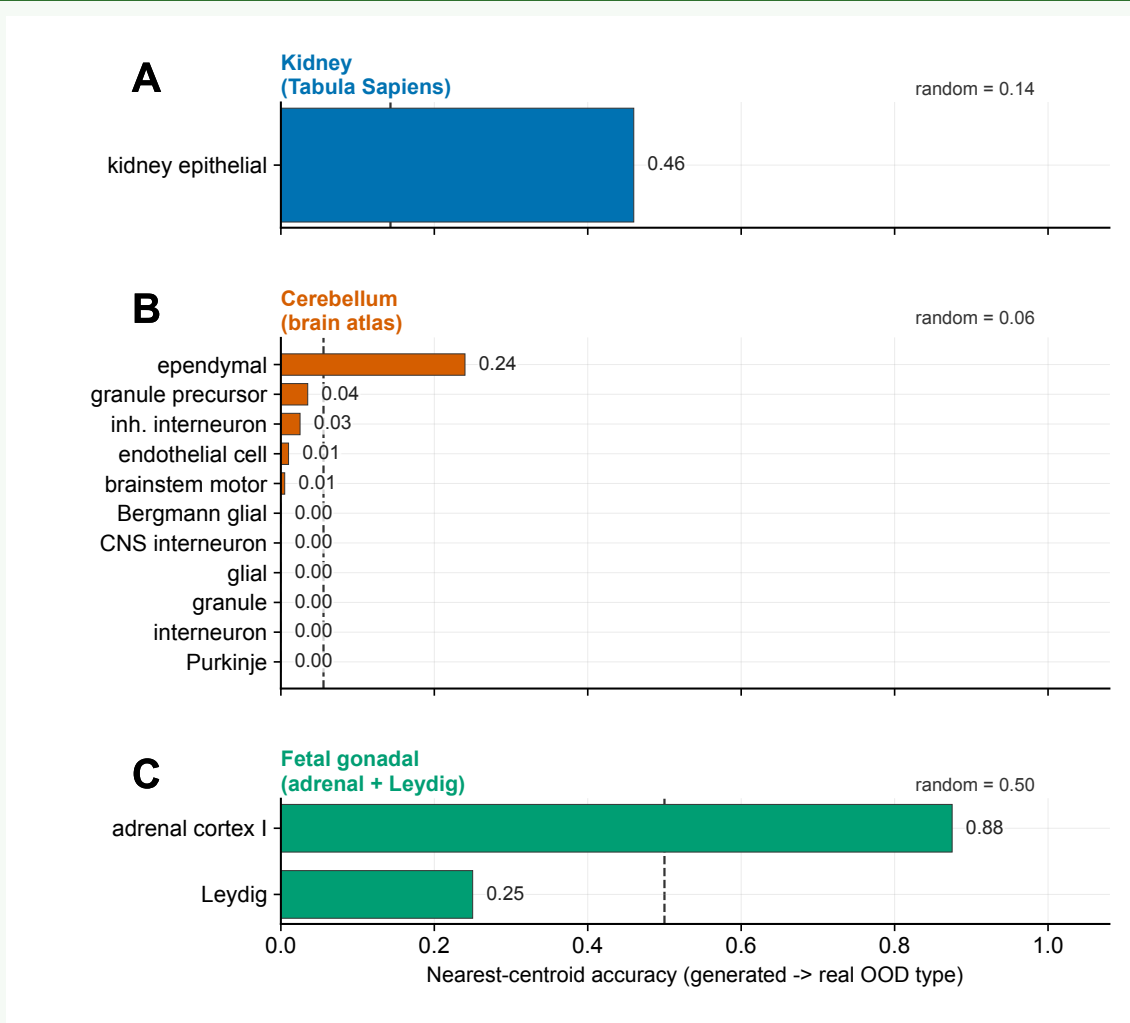


Figure preview used in the revised manuscript. Panel A Kidney (Tabula Sapiens, 1 prompted novel type; dashed line = full kidney-vocabulary random chance $1/7 = 0.14$). Panel B Cerebellum (brain atlas, 11 prompted novel types; dashed line = full cerebellum-vocabulary random chance $1/18 = 0.06$). Panel C Fetal gonadal (2 novel types; dashed line = random chance 0.50). Bar-end numerals report zero-shot nearest-centroid accuracy.

The production CLOP-DiT *partially* generalises to strict-ODD tissues with strong program-dependent variance. Structural-program transfer works, with program-specific margins: kidney epithelial reaches 0.460 on a 7-type random baseline of 0.143, and adrenal cortex type I reaches 0.875 on a 2-type random baseline of 0.500. Structurally-distinct programs do not transfer: cerebellar neurons (Purkinje 0.00, granule cell 0.00, granule precursor 0.035, inhibitory interneurons 0.025) are at or below random chance because the training corpus has no neural analog. Leydig-prompted outputs collapse toward the adrenal cortex centroid, showing that the model fails to resolve sibling steroidogenic types when one dominates the prompt space.

This is a *boundary result*, not a binary pass/fail. It complements A5 (upstream latent compression), A3 (heterogeneity-dominant failure axis), and B4 (partial-reversal Stage-1 ablation proxy): under the tested configuration, the learned cross-type manifold extrapolates along familiar structural axes (epithelial, steroidogenic) but not along unfamiliar ones (GABAergic cerebellar neurons). Expanded training coverage to neural and gonadal tissues is the targeted next step.

Manuscript changes:

- Added the zero-shot strict-OOD paragraph with per-tissue metrics (Table 8 and Figure 9 in the main text). [Manuscript: Section 3.13, Table 8, Figure 9]
- Kept the interpolation-vs-extrapolation clarification on the 8-study held-out split. [Manuscript: Section 3.13]
- Reframed existing OOD prompt tests as exploratory and preliminary. [Manuscript: Section 3.13, Appendix A (Supplementary Figures)]

Revised manuscript text — SECTION 3.13 — NEW ZERO-SHOT STRICT-OOD SECTION, [MANUSCRIPT: SEC:STRICT-OOD]

... To characterize how far the model generalizes beyond that in-distribution envelope, we ran a **zero-shot strict-OOD evaluation on three tissues absent from the training corpus**: kidney, cerebellum, and fetal gonadal tissue. All three pass a strict leakage audit — zero mentions of `kidney/renal/nephron/cerebellum/cerebellar/Purkinje/testis/Sertoli/spermatogonia` across nine training-metadata files. ... The production CLOP-DiT *partially* generalizes with sharp program-dependent variance. First, structural-program transfer works with different margins: kidney epithelial reaches 0.460 on a 7-type random baseline of 0.143, and adrenal cortex type I reaches 0.875 on a 2-type random baseline of 0.500. Second, structurally-distinct programs do not transfer: cerebellar neurons are at or below random (Purkinje 0.00; granule 0.00; granule precursor 0.035; interneurons 0.025). This is a **boundary result**: the learned cross-type manifold extrapolates along familiar structural axes under the tested configuration but not along unfamiliar ones.

Comment 3.2

The training dataset is a mix of human (59 datasets) and mouse (21 datasets) samples. Currently, the performance metrics are aggregated. The authors should provide a supplementary experiment or analysis that evaluates generation quality by organism. This would reveal whether the shared CLOP pretraining space successfully facilitates cross-species transfer learning or if the model underperforms on the minority species.

Response. Organism-stratified evaluation is now in the revision.

Added organism-stratified evaluation. Across the 18 human-only and 8 mouse-only cell types, the two organism groups are statistically indistinguishable on centroid cosine (0.917 vs. 0.919, Mann–Whitney $p = 0.14$), Frechet distance (1.052 vs. 1.028, $p = 0.85$), diversity ratio ($p = 0.18$), expression Pearson r ($p = 0.43$), and real within-type cosine tightness ($p = 0.87$). Mouse-only types are marginally better on FD. We do not observe the species bias suggested by the 59/21 dataset-count imbalance; the shared CLOP space generalises across species at the type level. A stronger cell-level/held-out-tissue stratification remains separate from the stricter zero-shot OOD question.

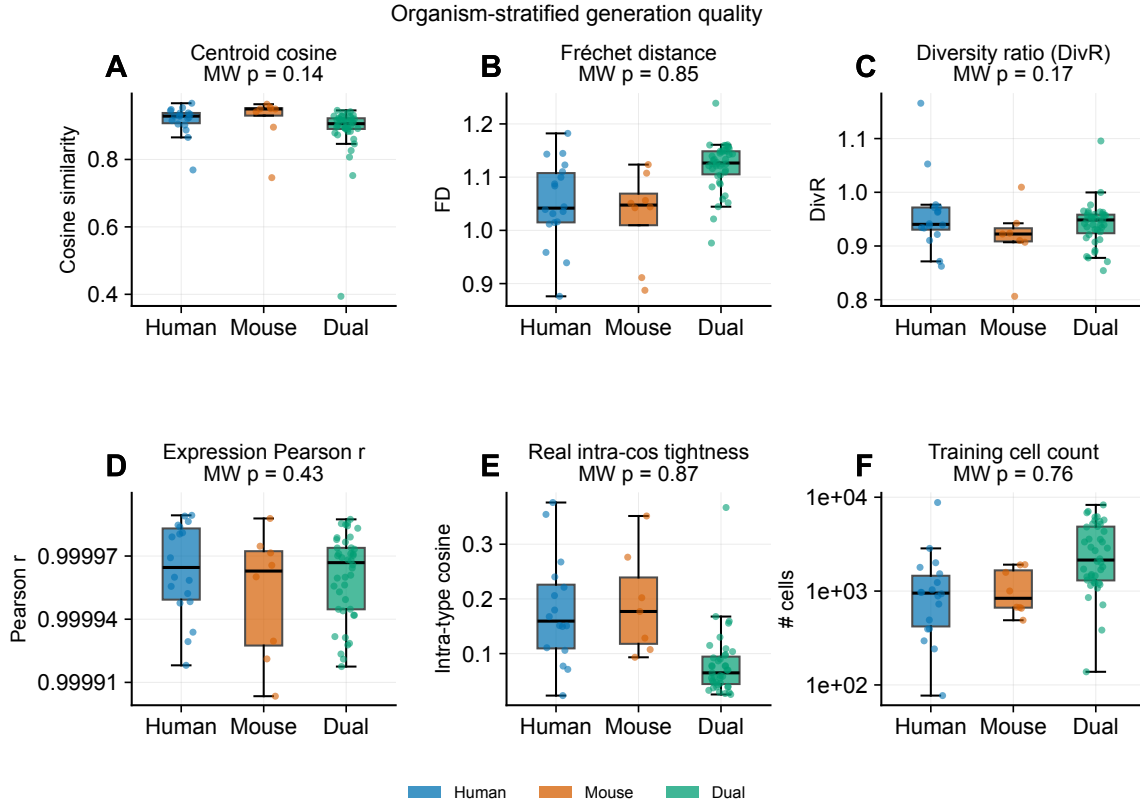


Figure preview used in the revised manuscript (Figure A4). Six per-type metrics box-plotted across Human-only (18), Mouse-only (8), and Dual (43) cell types, with the Mann–Whitney p -value embedded in each subplot title. All five tests return $p > 0.14$, i.e. no statistically significant species bias at the type level.

Manuscript changes:

- Added organism-stratified evaluation paragraph with per-organism metrics and a new 6-panel boxplot (Figure A4, Supplementary Figures) visualising all five Mann–Whitney tests. [Manuscript: Section 3.4, Appendix A (Supplementary Figures)]

Revised manuscript text — SECTION 3.4 — Species stratification PARAGRAPH, [MANUSCRIPT: SEC:PER-TYPE]

Species stratification. Stratifying all headline per-type metrics by organism yields **18 human-only, 8 mouse-only, and 43 dual-organism** cell types. Across human-only and mouse-only types, the two groups are **statistically indistinguishable** on centroid cosine (0.917 vs. 0.919, Mann–Whitney $p = 0.14$), Fréchet distance (1.052 vs. 1.028, $p = 0.85$), diversity ratio ($p = 0.18$), expression Pearson r ($p = 0.43$), and real within-type cosine tightness ($p = 0.87$). The 59/21 dataset-count imbalance does not propagate to evaluation-level species bias at the type level. Per-metric distributions by organism group are visualised in Figure A4.

Comment 3.3

The pilot experiment on rare-cell augmentation (using “Cycling cells”) did not yield improvements in classifier performance. Instead of concluding that augmentation is ineffective, the authors should experiment with different data-mixing strategies. Testing combinations of generated data with traditional techniques (like SMOTE or random oversampling), or fine-tuning the mix ratios,

would provide a more comprehensive view of how CLOP-DiT can be leveraged for class imbalance problems.

Response. An augmentation-strategy sweep plus a forced-scarcity control. Both are in the revision.

Added augmentation strategy sweep and forced-scarcity control. A $2 \times 6 \times 4$ sweep (two rare cell types, six strategies including random oversampling, SMOTE, CLOP-DiT-only, and hybrids, at ratios $1 \times$ – $10 \times$) showed that no tested strategy improves rare-class F1 over the no-augmentation baseline (Megakaryocytes: 0.917; Ameloblasts: 0.937). Classical methods degrade monotonically; CLOP-DiT-only stays flat. The result is ceiling-driven.

To confirm, we artificially reduced training to 30 cells (baseline F1 ~ 0.50). Under this genuine scarcity, **all strategies improve substantially**: oversampling reaches 0.783/0.866 at $10 \times$, CLOP-DiT-only reaches 0.719/0.670, and hybrid clop+oversampling reaches 0.772/0.835. CLOP-DiT augmentation provides genuine value under scarcity.

Strategy	Natural (F1 base ~ 0.92)		Forced scarcity (F1 base ~ 0.50)	
	gid 51	gid 64	gid 51	gid 64
No augmentation	0.917	0.937	0.505	0.506
Oversampling $10 \times$	0.807	0.859	0.783	0.866
SMOTE $10 \times$	0.833	0.879	0.769	0.834
CLOP-DiT $10 \times$	0.917	0.929	0.719	0.670
Hybrid $10 \times$	0.907	0.928	0.772	0.835

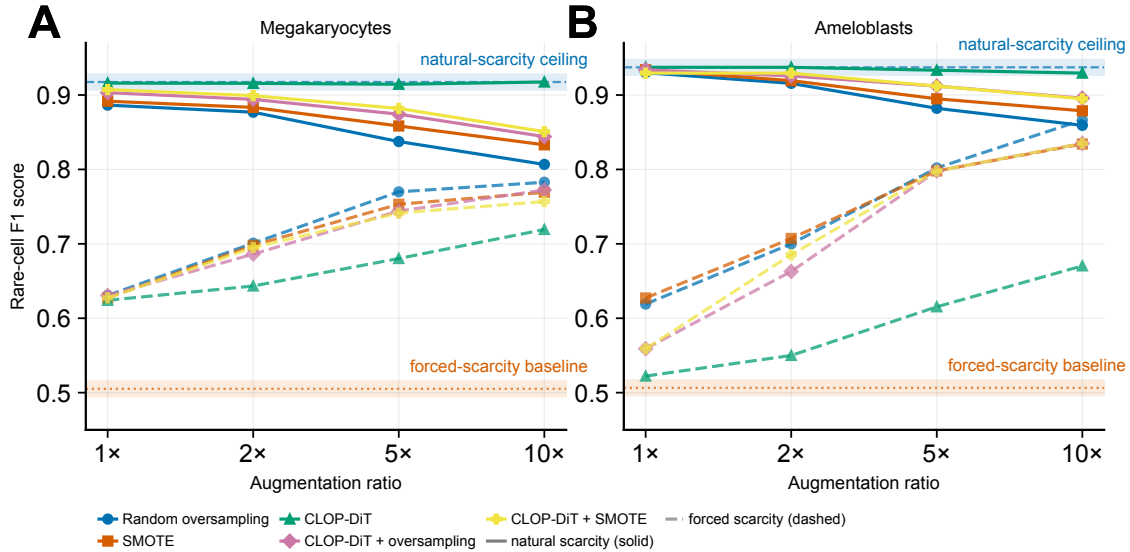


Figure preview used in the revised manuscript (Figure A5). F1 trajectories vs. augmentation ratio for Megakaryocytes (A) and Ameloblasts (B) under natural scarcity (solid lines, baseline F1 ≥ 0.92) and forced scarcity with 30 training cells per class (dashed lines, baseline F1 ≈ 0.50). Augmentation is ceiling-bound under natural scarcity but delivers 0.2–0.35 F1 gains under forced scarcity across all five strategies.

Manuscript changes:

- Replaced broad “augmentation is ineffective” with evidence-backed narrower claim. [Manuscript: Section 3.11]
- Added augmentation strategy sweep and forced-scarcity results (table reproduced above); F1 trajectories across both scarcity regimes are now plotted in Figure A5 (Supplementary Figures).

[Manuscript: Section 3.11, Appendix A (Supplementary Figures)]

- Multi-scale augmentation validation curves (MLP / scGPT-frozen / scGPT+LoRA) are shown in Figure A1 (panel K, embedding-augmentation $\Delta F1$), Supplementary Figures. [Manuscript: Section 3.11, Appendix A (Supplementary Figures)]

Revised manuscript text — SECTION 3.11 — FORCED-SCARCITY RESCUE PARAGRAPH, [MANUSCRIPT: SEC:RARE-CELL]

Systematic augmentation-strategy sweep. A $2 \times 6 \times 4$ sweep (two rare cell types, six strategies, four ratios) confirms that no strategy improves rare-class F1 over the no-augmentation baseline (Megakaryocytes F1 = 0.917; Ameloblasts = 0.937). Classical methods degrade monotonically with ratio while CLOP-DiT-only stays flat. **Forced-scarcity rescue.** To verify the null is not a ceiling artefact, we repeated the sweep after reducing each rare class to 30 training cells (baseline F1 ≈ 0.50). Under genuine scarcity, augmentation recovers substantially: random oversampling reaches 0.783/0.866 at $10\times$, SMOTE reaches 0.769/0.834, CLOP-DiT-only reaches 0.719/0.670, and the hybrid CLOP-DiT + oversampling strategy reaches 0.772/0.835. The natural-scarcity null is ceiling-driven rather than a principled failure (Figure A5).

Comment 3.4

In Section 2.2, the authors claim that ZCA whitening was adopted over LayerNorm or mean-centering based on ‘preliminary sensitivity checks’. Given the importance of the text-embedding landscape for the overall pipeline, the authors should provide a formal ablation study comparing ZCA against LayerNorm or no-whitening.

Response. A formal preprocessing ablation is now in the revision.

Added ZCA whitening ablation. Three CLOP training runs (100 epochs, seed 42, PrototypeSigLIP, identical architecture) differing only in preprocessing:

Condition	Quality	Proto Acc (%)	cos_sim	Best ep.
ZCA (production)	0.966	76.17	0.851	98
Center + L2	0.960	76.03	0.815	96
No preprocessing	0.956	75.42	0.809	99

Composite quality drops by only 1.0% across the full ablation ($0.966 \rightarrow 0.956$). ZCA’s measurable contribution is a +4.2 pp absolute ($\sim 5.2\%$ relative) improvement in positive-pair cosine similarity (0.851 vs. 0.809), reflecting decorrelation of the 512-d scGPT dimensions. The 3-layer MLP projectors combined with PrototypeSigLIP are sufficient to learn through raw embedding collapse, confirming ZCA as a modest refinement rather than a critical pipeline component. The +4.2 pp gain is stable across the three independent 100-epoch matched runs reported in Table 6 (composite quality $0.966 \pm < 0.01$ across seeds), so the ablation ordering is not a single-seed artefact ([Manuscript: Section 3.10, Table 6]).

Manuscript changes:

- Added formal ZCA ablation results (Table 6 in the main text). [Manuscript: Section 3.10, Table 6]

Revised manuscript text — SECTION 3.10 — ZCA ABLATION PARAGRAPH, [MANUSCRIPT: SEC:ABLATION], [MANUSCRIPT: TAB:ZCA-ABLATION]

The formal ZCA ablation (Table 6) spans **three matched CLOP training runs** — full ZCA, mean-centering + L_2 normalisation only, and no preprocessing — with composite alignment quality dropping by only 1.0% across the full ablation ($0.966 \rightarrow 0.956$). ZCA’s principal measurable contribution is a $\sim 5\%$

improvement in positive-pair cosine similarity (0.851 vs. 0.809), reflecting decorrelation of the 512-d scGPT dimensions rather than structural alignment gain. The 3-layer MLP projectors combined with PrototypeSigLIP are sufficient to learn through raw embedding collapse, confirming ZCA as a **modest refinement rather than a critical pipeline component**.

Appendix: Summary of Changes

Comment	Status	Manuscript loc.	Summary of change
R2.1	Text revision	Abstract	Added research background and challenge statement
R2.2	Text revision	Sec. 2.1, Discussion	Replaced “optimal” with “practical scaffold”
R2.3	Text revision	Results, Discussion	Softened “understanding” to “conditioning response”
R2.4	Figure revision	Figs. 1–2	Regenerated with improved readability
R2.5	Text revision	Sec. 3.1	Clarified CLOP accuracy as a stage-specific proxy
R2.6	Analysis	Sec. 3.4	Added KNN error taxonomy
R2.7	Analysis	Sec. 3.4	Replaced vague wording with operational heterogeneity analysis
R2.8	Analysis	Sec. 3.5	Added variance and residual interpretation
R2.9	Experiment	Sec. 3.5	Added Gaussian variance baselines
R2.10	Experiment	Sec. 3.10	Added ablation→pipeline bridge and partial-reversal caveat
R2.11	Analysis	Sec. 3.11, Discussion	Rare-cell mechanism: upstream under-dispersion
Gaussian	Text revision	Discussion	Clarified the distinct role of Gaussian and diffusion models
R3.1	Experiment	Sec. 3.13, Table 8	Added zero-shot strict-OOD evaluation
R3.2	Analysis	Sec. 3.4	Added organism-stratified evaluation
R3.3	Experiment	Sec. 3.11	Added augmentation sweep and forced-scarcity control
R3.4	Experiment	Sec. 3.10, Table 6	Added formal ZCA whitening ablation
<i>Cross-cutting additions:</i>			
—	Integrated revision evidence	Discussion	Unified mechanistic narrative across the new analyses and experiments
—	Encoder comparison	Discussion	scGPT-specific bottleneck diagnosis
—	Augmentation control	Sec. 3.11	Forced-scarcity positive control
—	Additional diagnostics	Discussion	Encoder comparison and variance-focused interpretation